جامعة أم القرى
UMM AL-QURA UNIVERSITY

A MACHINE LEARNING APPROACH FOR HEART DISEASE PREDICTION

A project submitted for the Data Analysis 1 course (DS2313)

COURSE INSTRUCTOR
(DR. Modhi Sharaf Al-shareef)

DURATION:
May 1 to June 6, 2026

SUBMITTED BY:
Reham Mohammed Abdoh
Lana Ahmed Alghamdi
Hanin Safar Alharthi
Lama Dhafer Al-qarni
Amnh Alahdal

DEPARTMENT OF DATA SCIENCE
COLLEGE OF COMPUTING
UMM AL-QURA UNIVERSITY

2026

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DATASET LINK

https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

DATES DATA ANALYSIS & MODELING

https://www.kaggle.com/code/reham487/machine-learning-project

ABSTRACT

This project applies machine learning techniques to predict heart disease using clinical and demographic patient data. The Heart Failure Prediction Dataset from Kaggle was analyzed to identify the most influential features associated with heart disease and to evaluate the performance of two classification models: Logistic Regression and Decision Tree Classifier. The dataset underwent several preprocessing steps, including exploratory data analysis, outlier handling using the IQR method, one-hot encoding for categorical variables, and feature scaling. Both models were trained and tested using an 80/20 split, and their performance was assessed using accuracy, precision, recall, and F1-score. The Decision Tree model achieved the highest performance across all metrics, demonstrating its suitability for this classification task. The findings highlight the potential of machine learning in supporting early detection of heart disease and improving clinical decision-making.

INTRODUCTION

The purpose of this project is to build a machine learning model capable of predicting heart failure based on clinical patient data. Heart disease is one of the leading causes of death worldwide, which makes early detection an important and impactful problem. The main objective of this study is to analyze the Heart Failure Prediction Dataset and identify the key factors that contribute to the likelihood of developing heart disease.

This project aims to answer the following research questions:

1. Which clinical features are most strongly associated with heart failure?
2. Can machine learning models accurately predict whether a patient has heart disease?
3. Which model provides the best performance for this classification task?

To address these questions, the dataset is explored, cleaned, and analyzed to understand its structure and characteristics. A machine learning approach is suitable for this problem because the dataset contains multiple clinical variables that can be used to classify patients into two categories: heart disease or no heart disease. Classification algorithms can learn patterns from the data and provide predictions that support medical decision-making.

METHODOLOGY

Data Source

• **Dataset** obtained from Kaggle
• **Title:** Heart Failure Prediction Dataset
• **File used:** heart.csv
• Loaded directly from the Kaggle environment using the correct directory path
• Contains clinical and demographic information related to heart disease

Dataset Description

• **Rows**: 918
• **Columns:** 12
• Represents patient records with clinical features
• First five rows displayed using df.head() to verify correct loading
• **Dataset includes two variable types:**
•**Numerical:** Age, RestingBP, Cholesterol,FastingBS, MaxHR, Oldpeak,HeartDisease
• **Categorical:** Sex, ChestPainType, RestingECG, ExerciseAngina, ST_Slope

Data Information

• Examined using df.info()
• All columns contain 918 non-null values
• No missing data
• **Data types include:**
• int64 (6 columns)
• float64 (1 column)
• object (5 categorical columns)

Statistical Summary

• Generated using df.describe()
• Provides key metrics: mean, minimum, maximum, quartiles
• Helps understand data distribution
• **Identifies unusual values** (e.g., Cholesterol ranges from 0 to 603)

EDA & Visualization

In this section, Exploratory Data Analysis (EDA) was performed to better understand the heart disease dataset before applying preprocessing and machine learning techniques. Several visualization methods were used to analyze data distributions and relationships between variables.

First, a count plot was created to visualize the distribution of patients with and without heart disease. The results showed that the dataset contains slightly more patients diagnosed with heart disease than healthy patients.

Next, a histogram with KDE was used to analyze the age distribution of patients. The visualization showed that most patients were concentrated in the middle-age and older-age groups.

A boxplot was also generated for the Cholesterol variable to detect outliers and examine the spread of cholesterol values. The plot revealed the presence of several extreme cholesterol values in the dataset.

In addition, a count plot was used to explore the distribution of chest pain types among patients. The ASY category appeared to be the most common chest pain type in the dataset.

Finally, a correlation heatmap was created to examine the relationships between numerical variables. The heatmap helped identify positive and negative correlations between features related to heart disease prediction.

Overall, the visualizations provided a clearer understanding of the dataset structure and supported the following preprocessing and modeling stages.

Data Preprocessing

In this phase, we prepared the dataset for machine learning modeling to ensure high accuracy and reliable results:

1. Checking for Missing Values

The dataset was examined for missing values using the *df.isnull().sum() method* in Pandas. The results showed that all columns contained zero missing values, indicating that the dataset is fully complete and does not require further preprocessing techniques, such as imputation or row removal. Ensuring the absence of missing values is a crucial preprocessing step, as incomplete data can negatively impact model training and lead to inaccurate predictions.

2. Checking and Handling Outliers

Outliers in the numerical features were identified using *Boxplots* to visualize the data distribution and detect extreme values. The initial plots revealed the presence of several outliers in continuous variables, specifically RestingBP, Cholesterol, and Oldpeak.

To handle these extreme values without losing valuable data, the Interquartile Range (IQR) technique combined with Capping (Winsorization) was applied. The first quartile (Q1) and third quartile (Q3) were calculated for each targeted feature to compute the IQR. Lower and upper thresholds were determined using the standard formulas:

- IQR = Q3 - Q1
- Lower Threshold = Q1 - 1.5 * IQR
- Upper Threshold = Q3 + 1.5 * IQR

Instead of removing rows, the extreme values falling outside these boundaries were capped using *the .clip()* method, bringing them within the acceptable theoretical range.

After preprocessing, the updated Boxplots demonstrated a much cleaner and more balanced distribution, with a complete elimination of visible outliers in RestingBP, Cholesterol, and Oldpeak. Since capping was used rather than row deletion, the dataset size remained completely unchanged, ensuring no information loss occurred.

Finally, variables such as FastingBS and HeartDisease are binary indicators (containing only 0 and 1). Consequently, they were excluded from this treatment, as

traditional outlier detection methods do not apply to categorical states represented numerically.

3. Checking Duplicate Rows

The dataset was evaluated for duplicate records using the *df.duplicated().sum()* method in Pandas. The output confirmed that there are zero duplicated rows across the dataset. This ensures that every record represents a unique observation, thereby enhancing overall data quality and preventing potential bias during model training caused by redundant samples.

4. Checking Categorical Variables

Categorical columns were analyzed using the *df[column].unique()* method in Pandas to identify all distinct categories within each feature. This step is essential to verify data consistency and detect any potential formatting anomalies, spelling errors, or unexpected values prior to the encoding phase.
The unique values for each categorical feature were identified as follows:
Sex: ['M', 'F']
ChestPainType: ['ATA', 'NAP', 'ASY', 'TA']
RestingECG: ['Normal', 'ST', 'LVH']
ExerciseAngina: ['N', 'Y']
ST_Slope: ['Up', 'Flat', 'Down']
The analysis confirmed that all categorical entries are clean, consistent, and well-formatted, making them fully prepared for feature encoding in the subsequent preprocessing stages.

• **One-Hot Encoding :** Since machine learning algorithms require numerical input, we used the pd.get_dummies() function to convert categorical variables into a binary format (0 or 1). By setting drop_first=True, we successfully avoided the multicollinearity issue, which improves the stability of our models.

[17]:

	Age	RestingBP	Cholesterol	FastingBS	MaxHR	Oldpeak	HeartDisease	Sex_M	ChestPainType_ATA	ChestPainType_NAP	ChestPainType_TA	RestingECG_Normal	RestingECG_ST
0	40	140	289	0	172	0.0	0	True	True	False	False	True	False
1	49	160	180	0	156	1.0	1	False	False	True	False	True	False
2	37	130	283	0	98	0.0	0	True	True	False	False	False	True
3	48	138	214	0	108	1.5	1	False	False	False	False	True	False
4	54	150	195	0	122	0.0	0	True	False	True	False	True	False

• **Feature Scaling:** To ensure that all numerical features contribute equally to the model's performance and to prevent features with larger scales from dominating, we applied the StandardScaler. We used fit_transform on the training set to learn

the parameters and transform on the test set to ensure consistency and prevent data leakage.

```
Training set shape: (560, 15)
Testing set shape: (141, 15)
```

Training Procedure :

This stage involves preparing the data to be fed into the selected machine learning models:

• **Data Splitting:** We partitioned the data into features (X) and the target variable (y), which represents the 'HeartDisease' status. The dataset was then split into a training set and a testing set using train_test_split with a ratio of 80% for training and 20% for testing (test_size=0.2). This split is essential to evaluate the model's ability to generalize on unseen data

```
تم تحجيم البيانات بنجاح.
```

Training and Prediction

In this part of the project, I trained two models: **Logistic Regression** and **Decision Tree Classifier**, to see which one performs better on the heart disease dataset.
Before training the Logistic Regression model, I needed to find the best number of iterations so the model could converge correctly. Instead of choosing a random number, I made a `while` loop that increases `max_iter` by 100 each time until the model runs without any warnings. This step wasn't required, but I added it because I thought it was a smarter way than guessing the number manually. The loop showed that the best value for `max_iter` is **1100**.
For the Decision Tree model, I also used a loop to find the best `max_depth`. I started from depth 1 and kept increasing it until the accuracy stopped improving. The best depth was **4**, and the accuracy was **0.875**.
After finding the best parameters, I trained both models using the training data. Then, I used them to predict the test data and printed the first 20 predictions from each model to take a quick look at the results.

	Model	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	0.853261	0.900000	0.841121	0.869565
1	Decision Tree	0.875000	0.903846	0.878505	0.890995

It is clear from the results that the two models are close in performance, but the decision tree is better in all metrics.

Training vs. Test Results

The models showed consistent performance between the training and test datasets. Logistic Regression achieved a training accuracy of **0.87** and a test accuracy of **0.85**, indicating good generalization with minimal overfitting. The Decision Tree model performed similarly, with a training accuracy of **0.88** and a test accuracy of **0.875**, showing stable behavior on unseen data. Overall, the close values between training and test accuracy confirm that both models learned effectively without memorizing the data.

```
Logistic Regression:
Training Accuracy: 0.8719346049046321
Test Accuracy: 0.8532608695652174
Decision Tree:
Training Accuracy: 0.8801089918256131
Test Accuracy: 0.875
```

RESULTS AND DISCUSSION

The two models were evaluated using four metrics: Accuracy, Precision, Recall, and F1-score. The results are summarized in the table below:

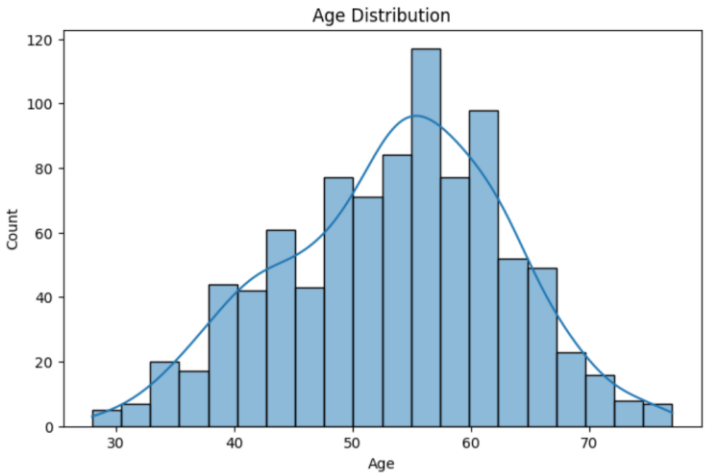
	Model	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	0.853261	0.900000	0.841121	0.869565
1	Decision Tree	0.875000	0.903846	0.878505	0.890995

After comparing the two models, I noticed that the Decision Tree performed better than Logistic Regression across all metrics. Its accuracy was 87.5% while Logistic Regression was 85.3%. What caught my attention the most is that the Decision Tree had a higher Recall (0.879 vs 0.841), which is very important, especially in the medical field, because high Recall means the model was better at detecting patients who are actually ill. In heart diseases, if the model fails to detect an infected case, the result could be serious, so the high Recall is considered an important advantage.

As for Precision, the two models were close to each other, but the Decision Tree was slightly higher (0.904 vs 0.900), which indicates that they rarely predict illness for a healthy person, and their results were accurate.

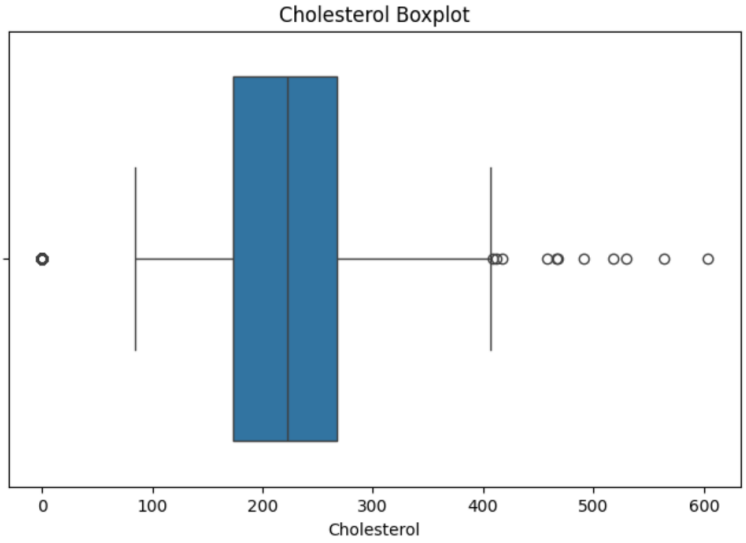
Overall, the Decision Tree was the better model for this data because it outperformed in all metrics.

Figure 1: Age Distribution



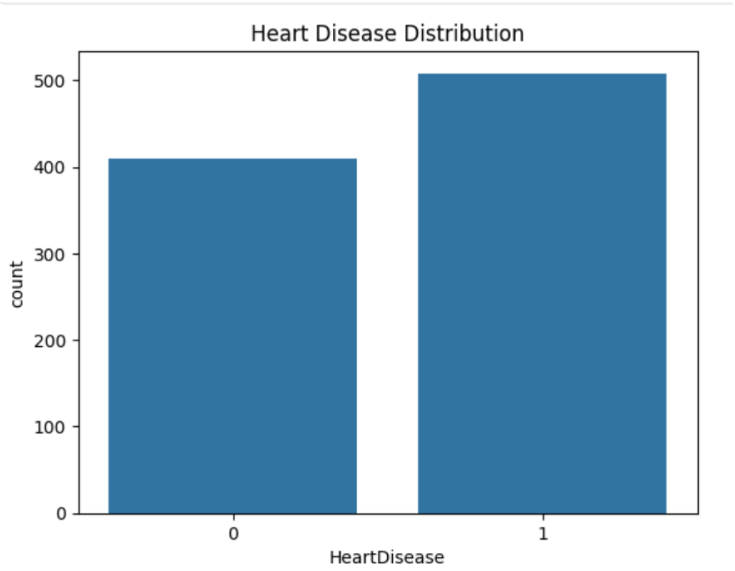
This figure shows the distribution of patient ages in the dataset. Most individuals fall between 45 and 60 years old, indicating that middle-aged patients are the majority in this study.

Figure 2: Cholesterol Boxplot



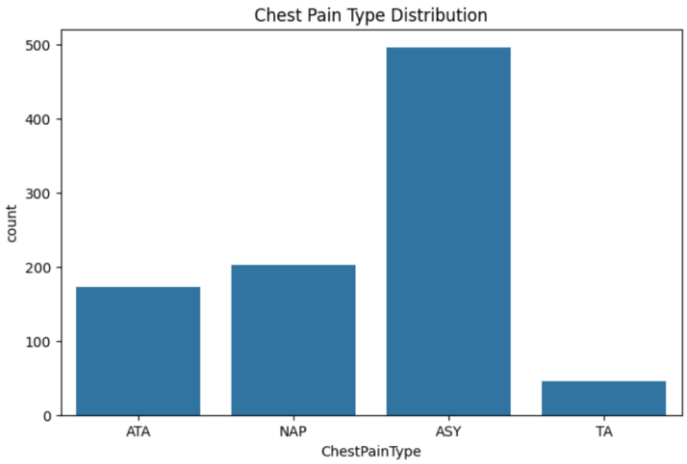
This figure shows the distribution of cholesterol levels among patients. Most values fall within a normal range, but several high outliers appear, indicating that some patients have unusually elevated cholesterol levels.

Figure 3: Heart Disease Distribution



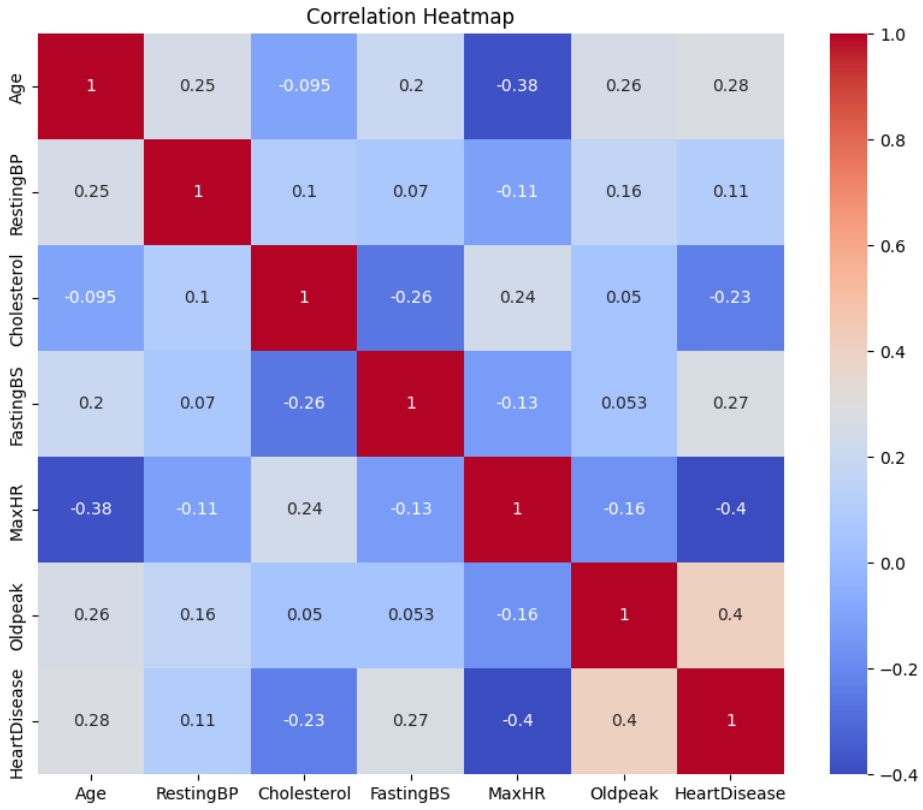
This figure shows the number of patients with and without heart disease. The dataset contains slightly more patients with heart disease than without, indicating a mild class imbalance.

Figure 4: Chest Pain Type Distribution



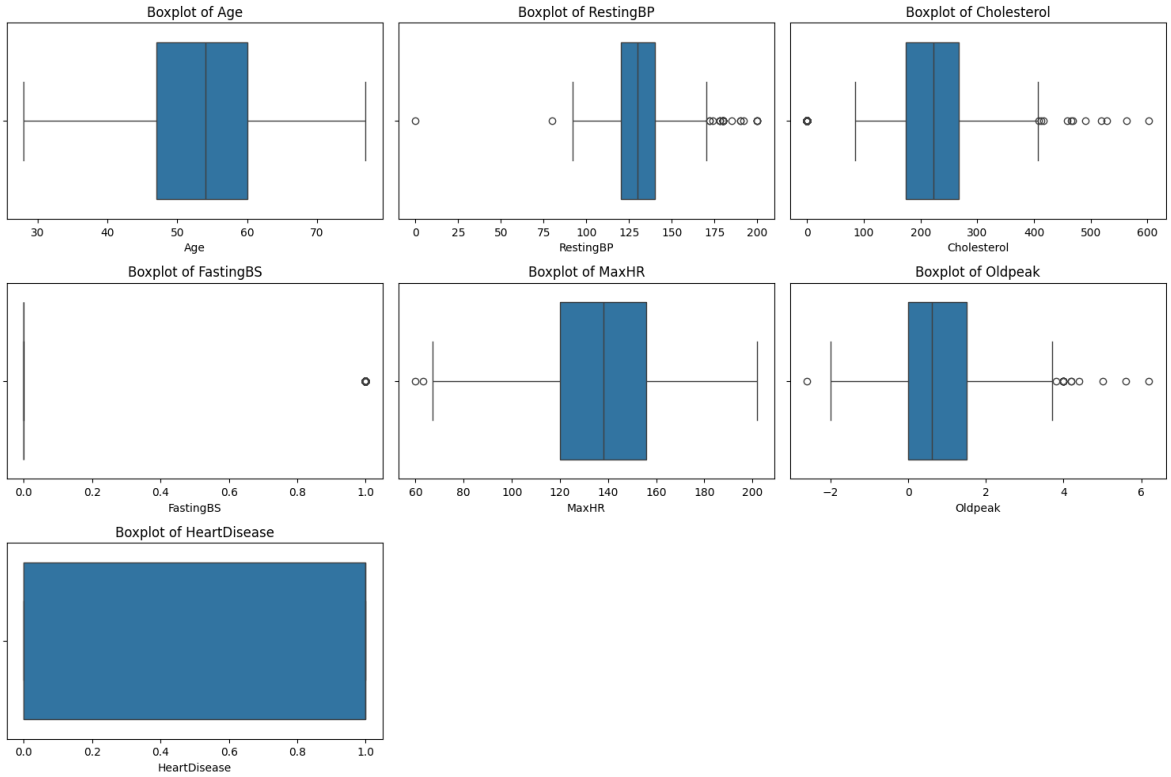
This figure shows the distribution of chest pain types among patients. The ASY type is the most common, followed by NAP and ATA, while TA is the least frequent.

Figure 5: Correlation Heatmap



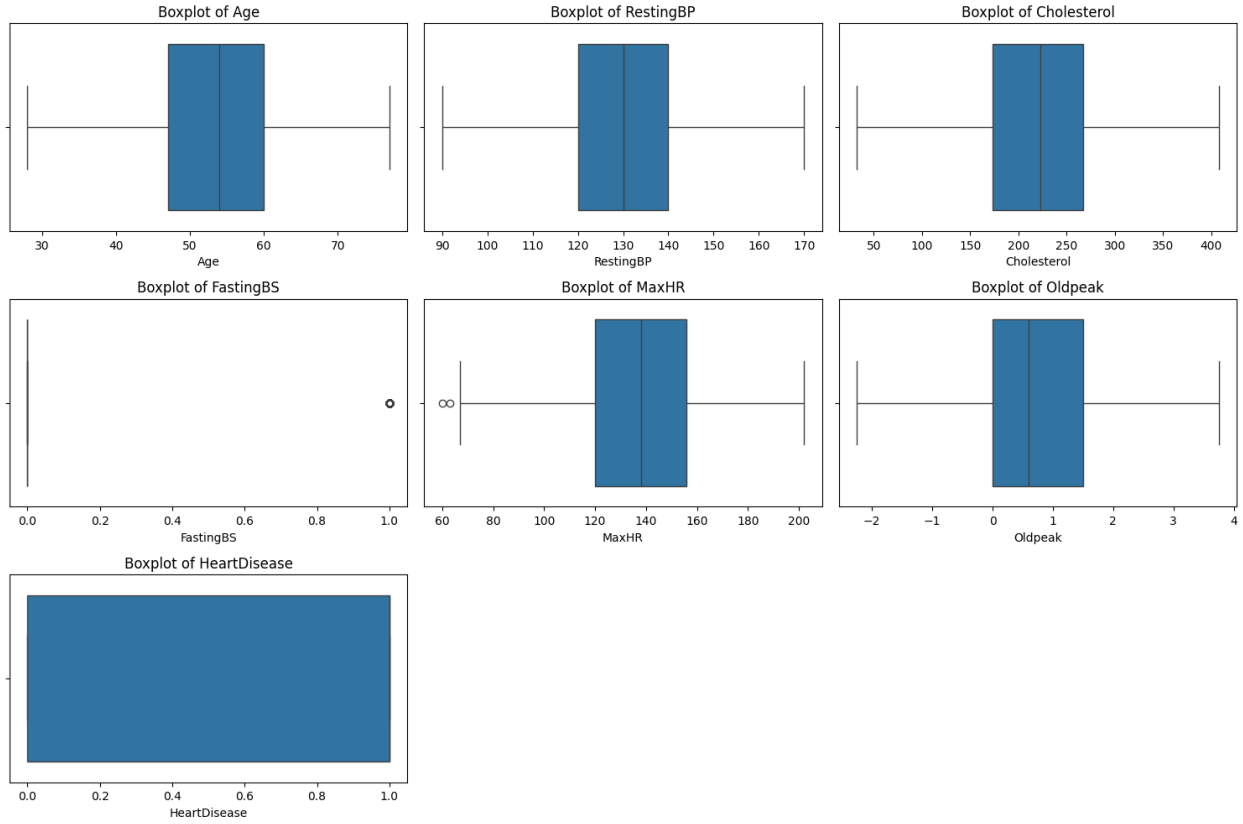
This figure shows the correlation between the main features in the dataset. Heart disease has a strong positive correlation with Oldpeak and moderate correlations with Age and Cholesterol, indicating that these variables are important predictors.

Figure 6: Outlier Detection Before Preprocessing



This figure illustrated the initial boxplots generated for the numerical features to detect potential outliers. Clear extreme values and anomalies were visible in continuous variables, specifically RestingBP, Cholesterol, and Oldpeak, which required treatment to ensure data quality.

Figure 7: Outlier Distribution After Preprocessing



This figure displayed the updated boxplots after the Interquartile Range (IQR) technique and capping via the .clip() method were applied. The visual results confirmed the complete elimination of extreme values in RestingBP, Cholesterol, and Oldpeak, which resulted in a cleaner and more stable distribution without any row deletion.

CONCLUSION

The aim of this project was to build a machine learning model capable of predicting heart diseases based on patients' clinical data. Two classification models were implemented and compared: logistic regression and decision tree classifier.

The results showed that both models performed well on the heart failure prediction dataset. However, the decision tree model achieved better results across all evaluation metrics, making it the preferred model for this classification task. The results indicate that machine learning can be an effective tool to support early detection of heart diseases.

Through this project, the team gained practical experience in the full machine learning cycle, including data exploration, preprocessing, model training, and evaluation. The project also highlighted the importance of selecting the appropriate model and tuning its parameters to achieve the best performance.

Despite the promising results, the project faced some challenges, particularly in handling outliers in continuous variables such as Cholesterol and RestingBP. For future work, it is recommended to explore additional algorithms such as Random Forest or Support Vector Machine, apply k-fold cross-validation for more robust evaluation, and collect larger datasets to improve model generalizability.

REFERENCES

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Sorianos, F. (2021). *Heart Failure Prediction Dataset*. Kaggle. https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

2. **Scikit-Learn Documentation**

Scikit-learn. (2024). *Machine Learning in Python: Documentation*. https://scikit-learn.org/stable/

3. **Course Materials**

Department of Data Science. (2026). *Data Analysis 1 (DS2313) — Course Materials*. Umm Al-Qura University.

THE EXPERIENCES AND SKILLS ACQUIRED BY THE TEAM MEMBERS

Student Name	Experiences	Skills
Reham	Conducting initial data exploration and describing the dataset.	Conducting initial data exploration and understanding dataset structure , Checking data quality, missing values, and data types , Describing dataset features and summarizing basic statistics ,Writing and organizing Section 1 of the report
Lana	Conducted exploratory data analysis and visualization for the heart disease dataset.	Using charts and graphs to analyze data , Analyzing age, cholesterol, and chest pain distributions , Detecting outliers and variable relationships , Contributing to writing Section 1 of the report.
Hanin	Conducted data cleaning and preprocessing for numerical and categorical features. Handled dataset anomalies and managed extreme values without information loss.	Checked data quality, missing values, and duplicate records , Validated categorical consistency and checked unique values ,Detected and visualized extreme values using Boxplots , Applied the Interquartile Range (IQR) technique and capped outliers.
Lama	Data Pipeline Preparation: Executing One-Hot Encoding, feature scaling, and data splitting (80/20) to ready the dataset for machine learning models.	ML Workflow Management: Proficient in data preprocessing techniques, standardization, and documentation of technical workflows.
Amnh	Comparing multiple machine learning models	Data cleaning and preprocessing .